FEATURE CONNECTIONS AND SIMILARITY MEASURES

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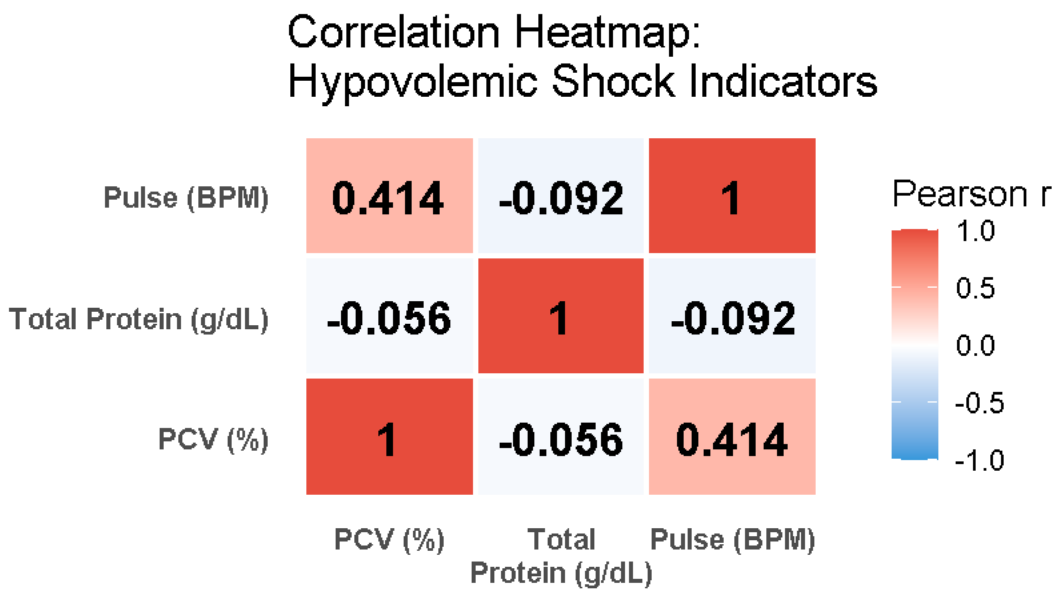
TASK 1 – CONNECTING PHYSIOLOGICAL MARKERS

- Selected features: Packed Cell Volume (PCV), Total protein, Pulse.
- Method: Pearson Correlation Matrix & Heatmap Visualization.
- Results:
  - Strongest connection: PCV and Pulse
  - Weakest connection: Total Protein and PCV



TASK 2 – DATA PREPROCESSING

- Goal: find the 2 most identical clinical cases in the dataset
- Subset selection: isolated 15 complete, NA-free clinical samples
- Dropped columns: "hospital_number", "nasogastric_reflux_ph", "abdomo_appearance", "abdomo_protein", "nasogastric_tube", "nasogastric_reflux", "rectal_exam_feces", "abdomen“
- Transformation for Euclidian math:
  - Categorical – dummy encoded to binary
  - Numerical – Z-score scaled

TASK 2 – RESULTS

- Method 1: Grower's distance
  - Row 3 and 43 are most similar (83,5% identical / distance of 0.165)
- Method 2: Euclidian distance
  - Row 1 and 14 most similar (distance 2.78)

```
> cat(">> METRIC 1: GOWER'S DISTANCE (Optimal for Mixed Data)\n")
>> METRIC 1: GOWER'S DISTANCE (Optimal for Mixed Data)
> cat(sprintf("   Most Similar Patient Pair : Row %s and Row %s\n", horse_A_gower,
horse_B_gower))
   Most Similar Patient Pair : Row 43 and Row 3
> cat(sprintf("   Gower Distance Score      : %.4f\n", min(gower_matrix, na.rm = T
RUE)))
   Gower Distance Score      : 0.1652
> cat(sprintf("   Overall Similarity        : %.2f%%\n\n", (1 - min(gower_matrix,
na.rm = TRUE)) * 100))
   Overall Similarity        : 83.48%

>
> cat(">> METRIC 2: EUCLIDEAN DISTANCE (Dummy Encoded & Scaled)\n")
>> METRIC 2: EUCLIDEAN DISTANCE (Dummy Encoded & Scaled)
> cat(sprintf("   Most Similar Patient Pair : Row %s and Row %s\n", horse_A_eucli
d, horse_B_euclid))
   Most Similar Patient Pair : Row 14 and Row 1
> cat(sprintf("   Euclidean Distance Score  : %.4f\n", min(euclid_matrix, na.rm =
TRUE)))
   Euclidean Distance Score  : 2.7879
```

CONCLUSIONS

- Task 1:
  - Successful identification of moderate positive relationship between PCV and Pulse
  - Blood thickening leads to increase in heart rate
- Task 2:
  - Different pairs selected using different methods